SPATIAL ECON | CITY OF EKURHULENI

ONBOARDING — APRIL 2026

The Ekurhuleni Spatial Economic Intelligence Platform.

A unified hexagonal model of Ekurhuleni's economy at 1.2 km resolution. Built on a national backbone covering all nine metros, nine provinces, and South Africa — the metro backbone is staged today, with eThekwini as the live, validated pilot. Tiers 1–3 already cover your geography from the national backbone — what's missing is your operational layer (Tier 4). This deck walks you through what we've built, what already runs for your geography on day one, and how a co-designed Tier 4 turns the platform into your operating system.

PREPARED FOR	City of Ekurhuleni Metropolitan Municipality
PREPARED BY	Spatial Economics
VERSION	v1.0 · 2026-04-29
STATUS	Outreach package — Ekurhuleni variant of the eThekwini pilot

PLATFORM AT A GLANCE

478
HEX-7 CELLS · EKURHULENI ESTIMATE

112
WARDS · CROSS-WALKED TO HEXES

19
INTEGRATED MODELS · M1–M19

32+
SOURCE FAMILIES · 1975 → FEB 2026

1000+
VARIABLES · SINGLE PANEL

500m
SPATIAL RESOLUTION · OUTPUT RASTER

HEXAGONAL ECONOMIC INTELLIGENCE — SUB-CITY, MONTHLY, VALIDATED.

THE THREE NUMBERS

If you remember nothing else from this hour, remember these.

§ 01 · FRAME

SPATIAL RESOLUTION

429 hex-7

Sub-ward, uniform geometry

H3 Level 7 hexagons — ~1.2 km edge — tile Ekurhuleni approximately 478 times; the same grid extends nationally as the staged scope is brought online. Wards are political; hexes are analytical. Every model, map, and forecast in the metro is keyed to this grid.

PRIMARY KEY · HEX7

TEMPORAL RESOLUTION

146 months

Monthly, 2014 → Feb 2026

Twelve full years of monthly observations on every hexagon. Long enough to see the COVID drop, the 2021 unrest, the 2022 floods, and load-shedding stages 4–6 — and recover from each.

COMPOSITE KEY · HEX7 × YEAR_MONTH

ANALYTICAL SURFACE

1B + rows

One panel. 1000+ variables.

Every signal — FTE, electricity, water, nightlights, faults, valuations, FDI projects, EDIP rebates, GHSL population, SANLC land cover, ward GDP, customs trade, plus all nineteen model outputs — sits on a single primary key. `hex7 × year_month` for the eThekwini ready slice; coarser geometry for the staged province and national scopes.

TABLE · HEX_PANEL

Sub-city, monthly, validated. Built once, queried hundreds of ways.

WHY HEXAGONS

Wards explain politics. Hexagons explain economics.

§ 01 · FRAME

Wards are uneven, politically defined, and re-drawn. Suburbs blur into each other. Pixel grids ignore neighbourhood structure. We needed a uniform analytical surface that respected geography but didn't inherit administrative noise.

H3 LEVEL 7

~1.2 km edge length

Each cell covers ~5.16 km². Smaller than a ward, larger than a property, comparable across the metro.

LOSSLESS AGGREGATION

Roll up, drill down

Hexes nest cleanly into wards (and out to the metro). Outputs reconcile to ward GDP via Denton-Cholette.

CROSS-METRO COMPARABLE

Same grid, any city

The same H3 spec runs the Western Cape Urban Growth Monitors. Ekurhuleni becomes one node in a national lattice.

EDGE-AWARE

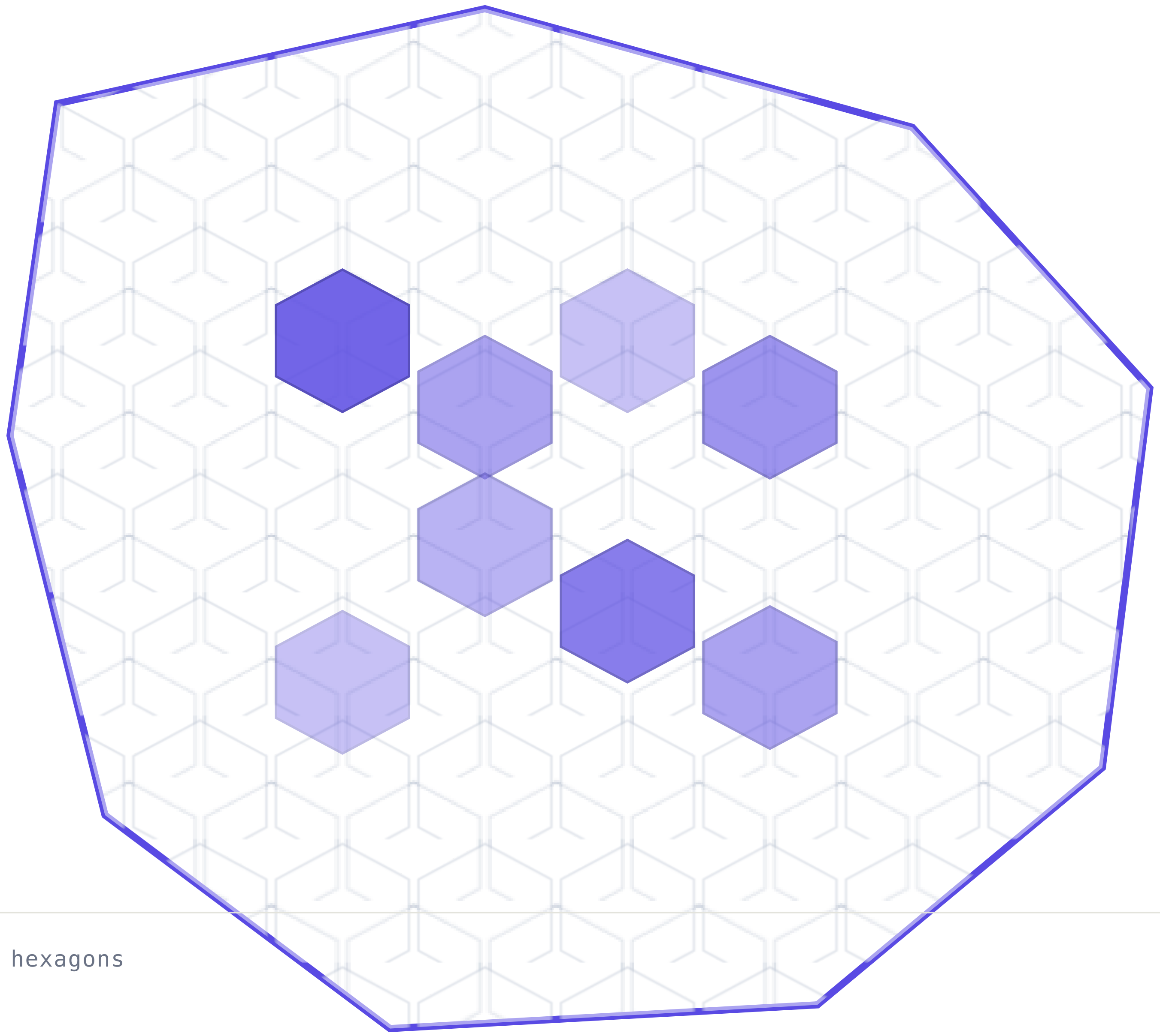
Economic activity crosses lines

A retail corridor can span three wards. Hexes capture the corridor; ward statistics dilute it.

GEOMETRY · SCHEMATIC

478 CELLS · EKURHULENI

Ward boundary vs H3-7 tiling



The data spine.

Thirty-plus governed sources. One key. Most municipalities have these in silos. We have them on a single primary key: `hex7 × year_month` for the Ekurhuleni metro slice; province / national scopes use coarser geometry from the same registry.

Thirty-plus feeds, four tiers, one key.

TIER 1 · GEOGRAPHY SPINE

Where things are.

SEADSA v6 — hex · muni · prov · nat
SPATIAL ECONOMIC ACCOUNTS
FTE · ESTABLISHMENTS · TOTAL_INCOME · GINI

GHSL — 100 m raster
EU JRC BUILT SURFACE · POP

SANLC — 20 m raster
SA NATIONAL LAND COVER

GIS estate — all levels
MOUNTED GEOMETRY LIBRARY
ROAD_LENGTH_M · RIVER_LENGTH_M · FLOOD_PLAIN_AREA_M2 · GEOHAZARD_AREA_M2

NATIONAL COVERAGE · READY

TIER 2 · STATISTICAL CONTEXT

How the country is doing.

StatsSA — ward → nat
CENSUS 2022, P3041, AFS, GFS, MONTHLY/QUARTERLY ECON
CPI · PPI · RETAIL_SALES · MINING_PROD · MANUF_PROD · POP_AGE_SEX · DWELLING_TYPE

QLFS — metro · prov · nat
QUARTERLY LABOUR FORCE
EMPLOYED · UNEMPLOYED · LABOUR_FORCE · NARROW_UNEMP · EXPANDED_UNEMP

S&P Metro GDP — 8 metros + nat
METRO_GDP_NOMINAL · METRO_GDP_REAL · SECTOR_GVA[19]

SARB — national
MACRO TIME SERIES
REPO_RATE · PRIME · ZAR_USD · M3 · CURRENT_ACCOUNT

NT Data — muni · nat
TREASURY MUNICIPAL FACTS + BUDGETS
OPEX_ZAR · CAPEX_ZAR · DEBT_SERVICE · OWN_REVENUE_SHARE

DataFirst — prov · nat
PUBLIC SURVEYS / MICRODATA
INCOME_DECILE · EXPENDITURE · SKILLS · MIGRATION

Provincial Data — muni · prov
KZN_TREASURY · PROVINCIAL_HEALTH · PROVINCIAL_EDU

NATIONAL COVERAGE · PARTIAL / READY

TIER 3 · FLOW & ACTIVITY

What's moving, when.

SARS Customs — postal → hex
HS·8 × COUNTRY × POSTAL/DISTRICT
EXPORT_VALUE_ZAR · IMPORT_VALUE_ZAR · TRADE_BALANCE · HS8_MIX

VIIRS nightlights — pixel → hex
ANNUAL NATIONAL PIXEL LATTICE
NIGHTLIGHT_SUM · NIGHTLIGHT_MEAN · RADIANCE_YOY_PCT

VIIRS Ward GDP — ward · metro
CALIBRATION TARGET FOR M1
WARD_GDP_ZAR · WARD_GDP_YOY · WARD_RADIANCE_IDX

Monthly satellite — ward → nat
NDVI · NDBI · NTL_MONTHLY · CLOUD_MASK_PCT

SAPS crime — station → nat
QUARTERLY STATION-LEVEL
CONTACT_CRIMES · PROPERTY_CRIMES · VIOLENT_IDX

NATIONAL COVERAGE · PARTIAL

TIER 4 · EKURHULENI OPERATIONAL LAYER · BRING YOUR OWN

Your data fills this tier.

Tiers 1–3 cover Ekurhuleni from the national backbone today. Tier 4 is the metro-or-province-private operational layer that turns the platform from a research surface into an operating system. For Ekurhuleni this would be your municipal billing, fault, valuation roll, incentive-rebate, fDi, firm-registry, council, and informal-settlement feeds.

Consumption elec_kwh · water_kl · charges_zar

Faults · M4 elec_fault_count · water_fault_count

Valuation rolls market_value_sum · property_count

Investment rebates rebate_zar_flow · jobs_flow · investment_zar

fDi Markets fdi_capital · fdi_jobs · fdi_projects_12m

Firm registry sic5 · sic7 · firm_count · sic_share

Council records sdbip_capex · decisions · minutes_topics

Informal settlements isp_area_m2 · isp_est_hhc

Other local feeds loadshedding · land use · industrial profiles

Co-design data with our team. The shapes above are the eThekwini reference; the schema is yours to define.

STORAGE

DuckDB · single file · 100 GB+

One persistent store. No second data lake. Spatial extension optional (toggle via env). Fully reproducible from `make all-local`.

Every source contributes named, typed columns to a single table.

A taste of what sits behind the source registry. These are real column names from `hex_panel` — the table every model reads from and writes back to. Grain is `hex7 × year_month`; rolling-12m derivatives carry the `_12m` suffix; ZAR fields end in `_zar`.

IDENTITY & PLACE

Keys, geography, environment

The fixed properties of a hex — what identifies it and what surrounds it.

hex7	H3-7 hex ID · primary key
year_month	YYYY-MM string
tax_year	SA fiscal year (Mar–Feb)
ward_no	Municipal ward 1–111
road_length_m	Road length intersecting hex
river_length_m	River length in hex
dam_area_m2	Dam polygon overlap
flood_plain_area_m2	100-yr flood plain area
geohazard_area_m2	Geohazard overlap
slip_failure_area_m2	Slip-failure zone overlap
urban_development_line_m	UDL length intersecting hex
isp_area_m2 · isp_est_hhc	Informal-settlement programme

ECONOMY & ACTIVITY

The economy + its monthly pulse

What the economy *is* (stock) and what's *happening* month to month (flow).

fte	Full-time-equivalent employment
establishments	Business establishments count
total_income	Total income (ZAR)
gini	Gini coefficient
market_value_sum	Sum property market values
property_count	Properties on the valuation roll
elec_kwh	Electricity consumption (kWh)
water_kl	Water consumption (kL)
elec_charges_zar · water_charges_zar	Utility charges (ZAR)
nightlight_sum · nightlight_mean	VIIRS pixel radiance · activity proxy
fault_count	All infrastructure-fault tickets
elec_fault_count · water_fault_count	Faults split by network · M4 inputs

INVESTMENT & OUTPUTS

Inflows + what the model writes back

Capital and projects flowing *in*, and the indices the M-series writes *back*.

fdi_capital_zar_flow	fDi Markets monthly capital flow
fdi_jobs_flow	fDi Markets jobs flow
fdi_projects_12m	fDi 12-month project sum
fdi_capital_real2020_zar_12m	Real-2020 12m capital sum
edip_rebate_zar_flow	Estimated monthly EDIP rebate
edip_jobs_flow	EDIP application-derived jobs
edip_investment_zar_flow	EDIP application-derived investment
gva_estimated	M1 · monthly hex GVA nowcast
eci · trade_eci	M2 · complexity · trade-complexity
accessibility_index	M3 · gravity job accessibility
resilience_score	M5 · resistance + recovery
data_quality_score	M0 · row-level coverage / freshness

CONVENTIONS Every column carries a **type**, a **source family**, and a **masking policy** for confidentiality (`<10` / `<100` SEADSA strings handled via `unmask_column()`). Around 1000 columns sit on the panel today; this slide shows roughly fifty representative ones grouped by their **role**: identity, activity, or output.

From raw data to decision surface in five stages.

STAGE 01

Ingest

Per-source loaders for SEADsa, VIIRS, consumption, valuation, faults, FDI, EDIP, GHSL, SANLC, Census.

`EKU INGEST`

STAGE 02

Harmonise

Spatial & temporal alignment to hex-7 × month. Raster zonal stats, suburb→hex fallback, masking.

`HARMONIZE/`

STAGE 03

Build panel

LEFT-JOIN every source onto the hex × month grid. Outputs `hex_panel` (1000+ vars).

`EKU BUILD-PANEL`

STAGE 04

Run models

Nineteen models (M1–M19) read from the panel and write back. Econometric shadow mode for forecasts.

`EKU RUN-MODELS`

STAGE 05

Validate & serve

Six programmatic checks. Dash dashboard, REST API, admin control plane, scenario lab.

VALIDATE · DASHBOARD · API

IDEMPOTENT

Re-runs are cheap

Every stage is checkpointed in DuckDB. Re-running a single stage doesn't invalidate downstream tables unless their inputs changed.

PORTABLE

Mac, Linux, Azure

Bootstrap creates the venv, installs extras, seeds env files. Doctor checks portability before deploy.

REPRODUCIBLE

One command rebuild

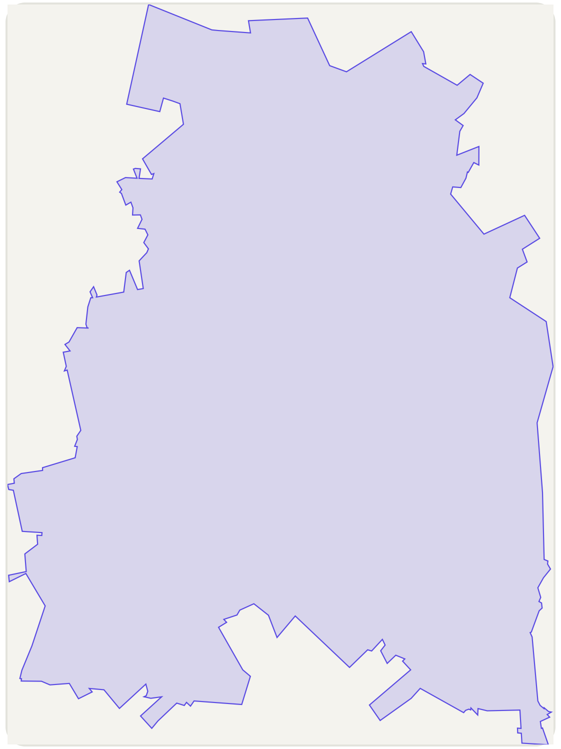
`make all-local` regenerates the panel and all nineteen models from the local data subset, no external volume required.

Five rungs from a national tax record to a 10 m pixel.

Most municipal economic data stops at the ward — a polygon containing tens of thousands of people. Our panel descends four further rungs, ending at a single 10 m land-cover pixel. Every rung below SEADSA is built; every rung is keyed to `hex7 × year_month` so the same query traverses all of them.

RUNG 01 · COARSEST

SEADSA municipal

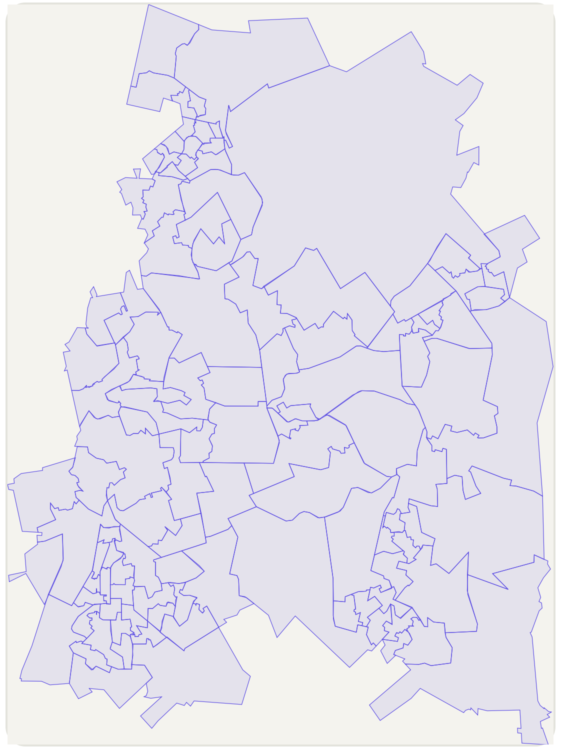


SOURCE	National Treasury · SARS
GEOMETRY	Municipality polygon
ROWS	1 / metro
UPDATES	Annual

Tax-record economic accounts. Authoritative, late, and one number for 4 million people.

RUNG 02

Ward

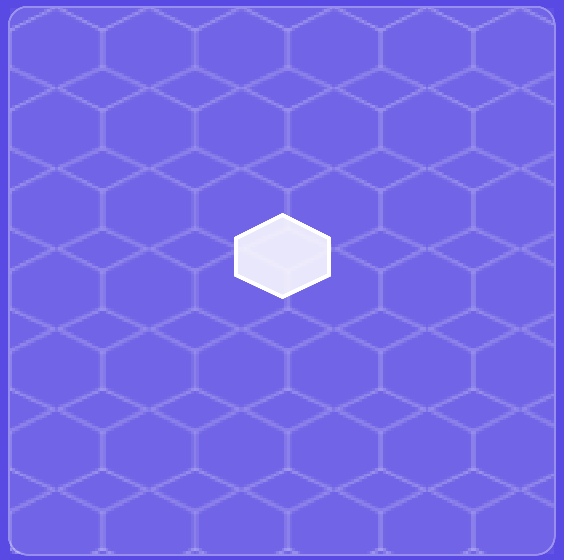


SOURCE	IEC + StatsSA
GEOMETRY	Political polygon
ROWS	~112 / metro
UPDATES	Per cycle

Where most public economic reporting stops. Boundaries follow politics, not economics.

RUNG 03 · OUR KEY

H3 hex (level 7)

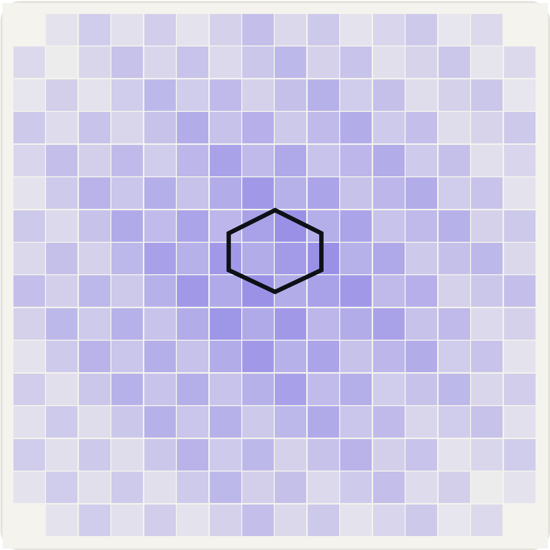


SOURCE	Uber H3
GEOMETRY	~5.16 km² hex
ROWS	429 / metro · ready
UPDATES	Monthly

Uniform, comparable, indifferent to ward redistricting. The single primary key for the panel.

RUNG 04

GHSL 100 m

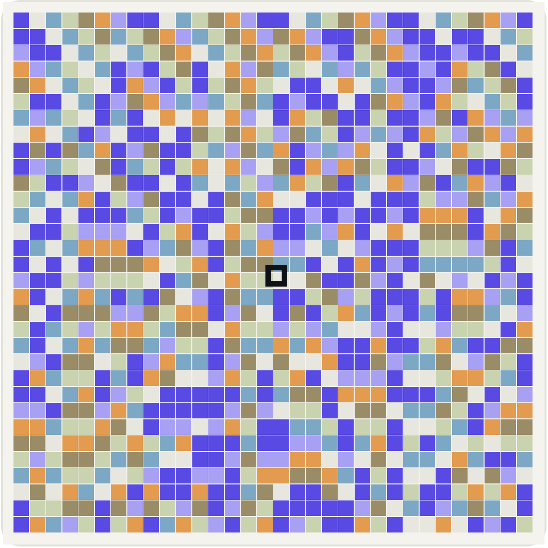


SOURCE	EU JRC GHSL
GEOMETRY	100 m raster cell
ROWS	~516 / hex
UPDATES	5-yearly

Built surface and population density inside the hex. Tells us where people actually live.

RUNG 05 · FINEST

SANLC 10 m



SOURCE	DFFE · SANLC 2022
GEOMETRY	10 m raster pixel
ROWS	~51 600 / hex
UPDATES	Periodic

73 land-cover classes. One pixel = one land-use decision. The atom of the model.

WHERE THE LADDER PAYS OFF

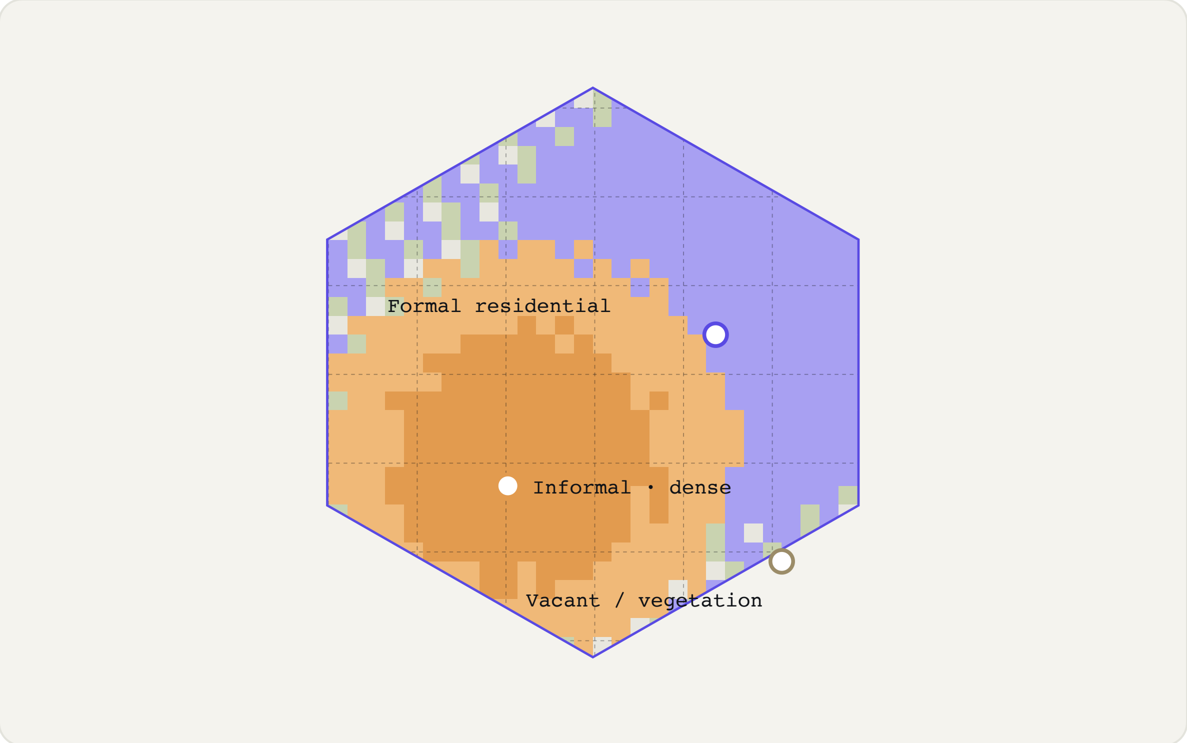
A hex where four sources disagree — and the disagreement is the signal.

§ 02 · DATA

One residential ward in the western metro. Inside it sits a single hex-7 cell containing roughly 18 000 people. SEADSA reports a coherent ward GVA. Inside the hex, the finer rungs disagree — and the disagreement is the formal/informal economy boundary, made measurable.

HEX 87BB1AA1D3FFFFF · WARD 102

Three layers, one cell.



SANLC 10 m mosaic clipped to hex · GHSL 100 m dashed overlay · each colour = land-cover class.

DISAGREEMENT MATRIX · THIS HEX, THIS MONTH

What each rung says about the same 5.16 km².

SOURCE	READING	WHAT IT IMPLIES	PICTURE
SEADSA · TAX-RECORD GVA	R 1.42 bn / yr	A respectable ward GVA, smoothed across 18 000 residents.	aggregate
Valuation roll · EDIP · CV2 16 YRS · REBATES	312 stands · 0 active	Formally-rated property covers ~38 % of built surface. No incentive footprint.	underweight
GHSL 100 m · POP DENSITY	14 200 ppl/km²	Population density 2.3× the ward mean. People are *here*.	dense
SANLC 10 m · CLASS MIX	61 % built · 22 % informal	22 % of built surface matches the informal-settlement signature.	flagged
VIIRS · NIGHTLIGHTS	+18 % YoY	Radiance rising faster than ward GVA. Activity is real, just not formal.	growing
Faults · M4 · TICKETS/MO	7.4 / km² avg	Fault density 3.1× metro mean — informal connections under stress.	stressed

THE READING ~4 000 informal households inside one ward GVA · invisible to municipal accounts · the disagreement is the formal/informal boundary made measurable.

Hex-level intelligence reveals what ward-level GDP cannot.

The ladder is not a technical curiosity. It is the difference between an economic statistic and an economic *instrument* — a number that helps the metro decide where, and what, and when.

RESOLUTION UNLOCKS · 01

The informal economy becomes measurable.

SEADSA cannot see four thousand informal households inside one ward. The ladder can — by triangulating GHSL density, SANLC class signatures, and VIIRS radiance against the absence of valuation-roll and EDIP records.

~22% of metro built surface tags as informal · invisible to ward GVA

RESOLUTION UNLOCKS · 02

Sub-ward inequality becomes visible.

One ward, one number averages the rich half and the poor half together. The ladder splits the same ward into 4–7 hexes, each with its own GVA estimate, accessibility score, and resilience trajectory. The Gini doesn't disappear into the mean.

7× finer than ward · ~5.16 km² per cell · monthly cadence

RESOLUTION UNLOCKS · 03

Infrastructure failure localises to economic loss.

A fault in one hex shows up as a measurable GVA dip in the same hex (M4). At ward scale this is noise. At hex scale it is a coefficient — and a budget argument the metro can defend.

1 hex is the unit at which fault → GVA causality is identifiable

THE THESIS IN ONE LINE

Wards tell you that the economy moved. The ladder tells you where, why, and to whom.

Nineteen models. One panel.

M1 nowcasts the economy month by month. M2 measures complexity. M3 quantifies the apartheid spatial legacy. M4 links infrastructure to GVA. M5 measures resilience to named shocks. M6–M19 extend coverage to property, growth, land, trade, labour, climate, firms, fiscal health, inclusion, informal economy, and governance — every one of them reads from `hex_panel` and writes back into it.

Each model answers one question. All nineteen share the panel.

M1 · NOWCASTING

Monthly hex GVA

Proxy regression on FTE, electricity, water, nightlights — calibrated to ward GDP via Denton-Cholette.

→ gva_estimated

M2 · COMPLEXITY

Hidalgo–Hausmann ECI

Revealed comparative advantage and complexity at intra-city scale. First African application.

→ eci · diversification

M3 · ACCESSIBILITY

Gravity job access

The apartheid spatial legacy as one index per hex. Gradient across former group areas is the headline.

→ accessibility_index

M4 · INFRASTRUCTURE

Fault → GVA

Panel regression of fault tickets and repair hours against GVA. Causal coefficient per fault hour.

→ predicted_gva_loss

M5 · RESILIENCE

Shock recovery

COVID, KZN floods, July 2021 unrest, load-shedding. Resistance + recovery months per hex.

→ resilience_score

M6 · HEDONIC

16-year property model

Hedonic regression on the valuation roll — value vs amenity, access, infrastructure.

→ value_premium

M7 · URBAN GROWTH

50-year typology

GHSL built-surface change (1975–2030) classified into growth typologies per hex.

→ growth_typology

M8 · LAND ECONOMY

SANLC transitions

Dominant land-use classification linked to economic outcomes (cropland → built, etc.).

→ dominant_landuse

M9 · TRADE

Customs gravity

SARS hex×SIC×country trade rolled into dependency and supply-chain risk per hex.

→ trade_dependency · supply_chain_risk

M10 · LABOUR

Spatial mismatch

Where the jobs are vs where the workers live. Mismatch + exclusion scores per hex.

→ spatial_mismatch · exclusion

M11 · CLIMATE

Hazard exposure

Flood, heat, sea-level risk overlaid with economic activity. Adaptation priority ranking.

→ climate_risk · adaptation_priority

M12 · CGE / SAM

Sectoral SAM

SEAD-SA-derived social accounting matrix. Sector ⇄ sector multipliers feeding scenario runs.

→ sectoral multipliers

M13 · FIRMS

Firm dynamics

Birth, death, and entry rates by hex. Where the startup ecosystems are forming.

→ startup_ecosystem · entry_rate

M14 · COMPOSITE

Index rollups

Cross-model composite indices for dashboard layers and the pilot package.

→ composite layers

M15 · PULSE

Activity nowcast

High-frequency pulse from VIIRS + utilities. Anomaly flags when the city is off-trend.

→ activity_index · anomaly_flag

M16 · FISCAL

Municipal health

Council SDBIP and rates base. Liquidity, revenue mix, and capex execution at municipal level.

→ fiscal scorecard

M17 · INEQUALITY

Inclusion index

Income, access, and service distribution across hexes. Who the economy is leaving behind.

→ inclusion_index

M18 · INFORMAL

Informal economy

Estimates the informal share per hex and the formalisation potential.

→ informal_share · formalisation

M19 · GOVERNANCE

Service delivery

Faults, SDBIP, and council ops fused into a governance quality score.

→ governance_quality · service_trend

DEEP-DIVE →

Three of nineteen

The next three slides go deep on M1 nowcasting, M3 accessibility, M5 resilience — then a tour of the other sixteen.

M1 · NOWCASTING

Municipal GDPNow — every hex, every month.

National GDP arrives quarterly with a six-week lag. Our nowcaster produces a hex-level GVA estimate every month, calibrated to the ward GDP pipeline so totals reconcile.

METHOD

Proxy regression → Denton-Cholette

Step 1 — fit log-GVA against FTE, electricity, water, and nightlight intensity at the hex level on a calibration window. Step 2 — Denton-Cholette temporal disaggregation reconciles the monthly hex series back to the ward-quarter GDP target.

INPUTS

fte · elec_kwh · water_kl · nightlight_mean

OUTPUT

gva_estimated (R · monthly · per hex)

CALIBRATION

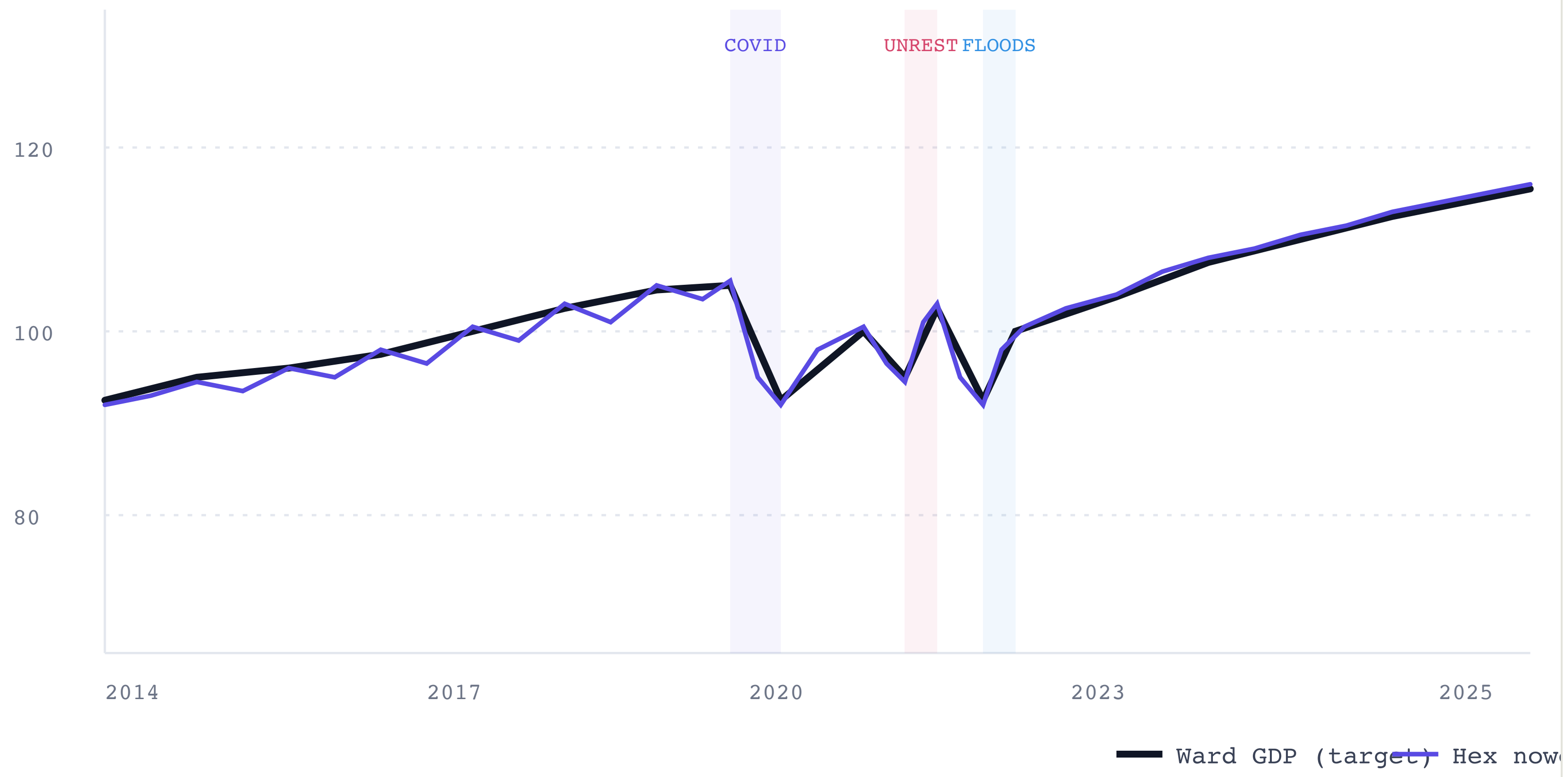
Ward GDP pipeline 2014–Feb 2026

VALIDATION

∑hex_gva ≈ ward_gdp
r > 0.95 (target)

ILLUSTRATIVE · MONTHLY HEX-LEVEL GVA (BEREA WARD SAMPLE)

Nowcast vs ward target



ILLUSTRATIVE CHART BASED ON THE MODELLED RELATIONSHIP — REPLACE WITH REAL WARD OUTPUT ONCE EXPORTED.

M3 · ACCESSIBILITY

A single number for the apartheid spatial legacy.

A gravity-based job accessibility index per hex — the weighted sum of every other hex's employment count, discounted by travel time. Numbers are interesting; the gradient across former group areas is the headline.

PULL QUOTE

> Two hexes 14 km apart. One sits at index 0.84 — within reach of half the metro's employment. The other sits at 0.21. The line between them was drawn in 1950.

METHOD	Gravity model · accessibility = Σ_j employment_j · f(travel_time_ij)
DECAY FUNCTION	Negative exponential, β calibrated against revealed commute distances
INPUT NETWORK	OSM road network · public transport schedules where available
OUTPUT RANGE	0–1 normalised across the 429 hexes

ILLUSTRATIVE · ACCESSIBILITY DECILES

Sample hexes ranked low → high

D10 · CBD	0.84
D9 · BEREA	0.78
D8 · UMHLANGA	0.71
D7 · PINETOWN	0.59
D5 · PHOENIX	0.43
D3 · INANDA	0.28
D2 · KWAMASHU	0.21
D1 · UMZINTO	0.12

ILLUSTRATIVE RANKING — REPLACE WITH THE ACTUAL HEX EXPORT FROM ACCESSIBILITY_INDEX.

M5 · RESILIENCE

Three named shocks. Resistance and recovery, per hex.

SHOCK 01

COVID-19

Lockdown peak: Q2 2020. Differential resistance varied by sector mix; hospitality hexes dropped >30%, light-industrial dropped ~12%.

WINDOW · 2020-03 → 2021-12

SHOCK 02

July 2021 unrest

Concentrated, severe, recoverable. Retail and warehousing hexes along the N3 corridor took the largest hit.

WINDOW · 2021-07 → 2022-06

SHOCK 03

KZN floods · 2022

Spatially concentrated along low-lying corridors. Infrastructure damage drives a multi-month tail in faults and recovery.

WINDOW · 2022-04 → 2023-03

DEFINITION

Resistance

The depth of the GVA drop relative to the pre-shock baseline. Measured as `1 − min(gva_t)/baseline`. Lower is better.

0.18

MEDIAN RESISTANCE · POST-COVID HEXES

0.34

WORST-DECILE RESISTANCE

DEFINITION

Recovery

Months to return to within ±2% of the pre-shock baseline. Measured per hex per shock; varies dramatically across the metro.

11 mo

MEDIAN RECOVERY · CBD HEXES

21 mo

MEDIAN RECOVERY · LOW-ACCESSIBILITY HEXES

NUMBERS ABOVE ARE ILLUSTRATIVE DEFAULTS. REPLACE WITH HEX_PANEL EXPORTS ONCE PRODUCTION VALIDATION PASSES.

M6 · M7 · M8 · M9 · M10 · M11 · M12 · M13 · M14 · M15 · M16 · M17 · M18 · M19

The other sixteen — grouped by what they answer.

PLACE & GROWTH

How the city is built and changing.

M6 Hedonic property — 16-year valuation roll → value_premium

M7 Urban growth — 50-year GHSL → growth_typology

M8 Land economy — SANLC → dominant_landuse

M14 Composite — cross-model index rollups

FLOWS & MARKETS

What moves through the city.

M9 Trade gravity — SARS hex × SIC × country → trade_dependency

M12 CGE / SAM — sector multipliers driving scenario runs

M13 Firm dynamics — birth, death, entry → startup_ecosystem

M18 Informal economy — share per hex → informal_share

SHOCKS & SPILLOVERS

What knocks the economy off-trend.

M10 Labour mismatch — jobs vs workers → spatial_mismatch

M11 Climate risk — flood, heat, sea-level → climate_risk

M15 Pulse — VIIRS+utilities anomaly flags → activity_index

DISTRIBUTION & INSTITUTIONS

Who is being served, and how well.

M16 Fiscal health — SDBIP, rates base → municipal scorecard

M17 Inclusion — income / service distribution → inclusion_index

M19 Governance — faults + ops fused → governance_quality

All sixteen read from `hex_panel` and write back into it. M12 SAM, M14 composite, and M16 fiscal write municipality-scoped rollups instead of hex columns — they consume the panel rather than extend it.

VALIDATION

Six checks. Run on every build. Fail loud.

WHY PROGRAMMATIC

Face validity, automated

"Industrial-zone hexes have higher ECI" should be true every build. We don't trust ourselves to remember to check — the test harness does it on every model run, in CI, and blocks deploy on regression.

TEST SURFACE

1,059 tests · 6 cross-model checks

Unit tests for utilities and individual models, integration tests for the pipeline, and a validation suite that exercises the cross-model relationships specifically.

#	CHECK	METHOD	TOLERANCE
V1	**M1 aggregation** HEX NOWCASTS ROLL UP TO WARD GDP	∑hex_gva by ward, correlate against ward-GDP target series	$r > 0.95$
V2	**M2 face validity** INDUSTRIAL HEXES HAVE HIGHER ECI	Compare ECI distribution in industrial-zoned hexes vs metro median	$\Delta > 1\sigma$
V3	**M3 income correlation** ACCESSIBILITY TRACKS EMPLOYMENT	accessibility_index vs FTE per hex	$r > 0.55$
V4	**M5 shock visibility** UNREST MONTHS SHOW MEASURABLE GVA DROP	July 2021 vs same-month-prior-year, paired hex test	drop ≥ 5%
V5	**M7 typology diversity** GROWTH CLASSIFICATION PRODUCES ≥3 DISTINCT TYPES	Distinct growth typology values across 429 hexes	≥ 3
V6	**Cross-model consistency** GVA CORRELATES WITH ECI AND ACCESSIBILITY	Pairwise correlation matrix across primary outputs	$r > 0.4$

What you'll touch.

The dashboard, the Econometric Lab, and the API are the surfaces you'll use day-to-day. Everything else is plumbing.

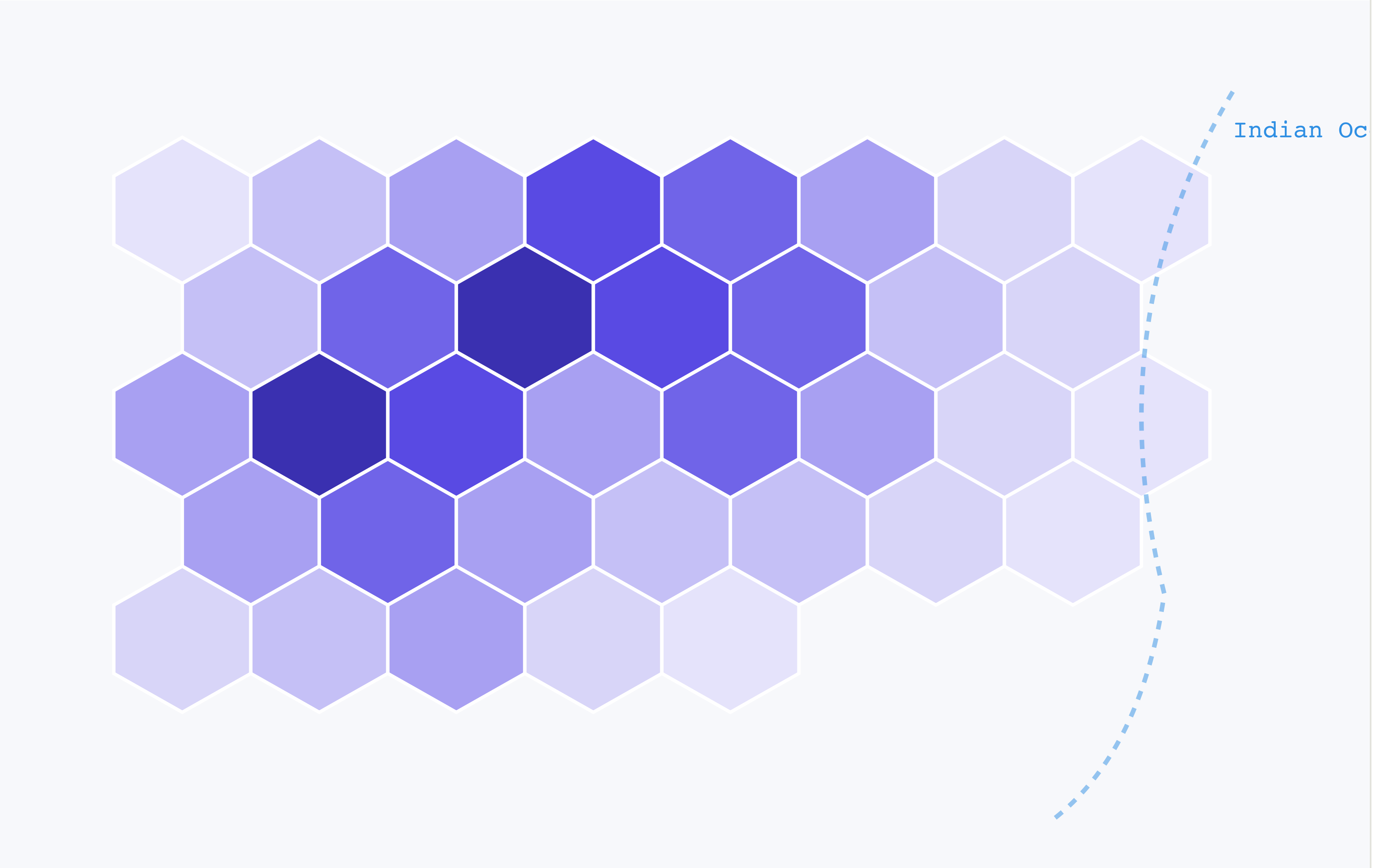
THE DASHBOARD

Every column on the panel. 146 months. Two clicks to the answer.

§ 04 · SURFACE

Hex choropleth · GVA · 2025-Q3

MOCK



GVA / HEX (R, MONTHLY) 0.5–1.5M 1.5–4M 4–9M 9M+

VARIABLE SELECTOR

Every panel column.

FTE · GVA · ECI · accessibility · resilience · property values · nightlights · growth rate · dominant land use · faults · CAPEX · EDIP · FDI flow · population · built surface · infrastructure quality · service-delivery urgency.

TIME CONTROL

Monthly slider · 2014 → Feb 2026

Drag through 146 months. The hex map and click-detail time series animate together.

COMPARISON

Side-by-side maps

Pin two variables for the same month, or pin one variable across two months. Directly answers "what changed".

AGGREGATION

Ward toggle

Flip from hex-level to ward-average for executive views. Underlying data unchanged.

Ask in English. Get a sourced answer with charts and caveats.

A Claude-powered `ChatAgent` sits over the same DuckDB store as the dashboard. It is read-only, scope-aware, and instrumented — every answer carries the SQL it ran, the tools it called, the freshness of each source, and the caveats the data quality registry attached. It is not a chat skin; it is the model's tool-use surface, exposed.

HOW A TURN RUNS

Three phases · one tool-use loop

1. **Pre-setup.** Resolve user role and scope, pick the workflow pack, build the system prompt, pre-load context.
2. **Tool loop.** Anthropic tool-use iterates: call a tool, read the result, decide the next call, terminate when it has the answer or hits the iteration cap.
3. **Post-synthesise.** Assemble Markdown answer + Plotly / MapLibre / Mermaid artefacts + caveats + source list. Persist to the conversation store.

TERMINATION CONTRACT

```
tool_loop_terminated_reason ∈ {
  "final",
  "max_tokens",
  "user_cancel",
  "tool_error"
}
```

Analyst transcript · sample

MOCK

```
You ▸
Which 5 wards have rising GVA momentum and elevated service pressure?

Analyst ▸ thinking…
[tool: gdp_momentum scope=ward limit=20]
[tool: service_pressure scope=ward]
[tool: rank metric=composite]

Top 5: W035, W098, W018, W067, W110.
Momentum +3.2 to +5.1 ppt YoY; pressure 0.71–0.84.

⚠ Caveats: M1 nowcast confidence 0.82 for W110.
[Tools: gdp_momentum, service_pressure, rank]
```

READ-ONLY

Agent uses a read-only DuckDB handle. No mutation tools.

SQL GUARD

Every query routed through `sql_guard`: allow-listed tables, row caps, plan inspection.

REPLAY

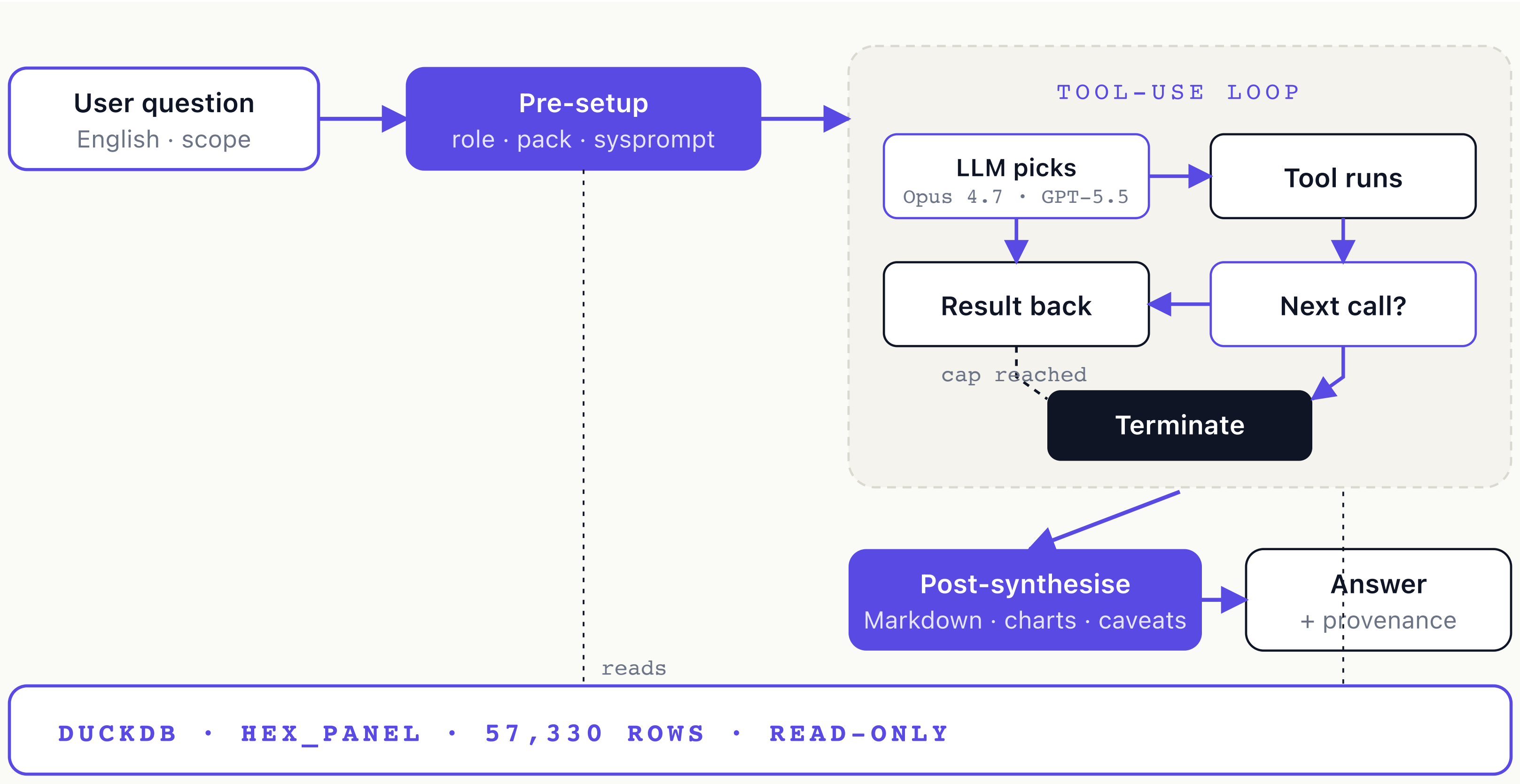
Every turn captured for replay + golden-output diffing across model versions.

The model is a panel. The AI is a tool-user.
It doesn't reason about the city — it queries it.

Our AI is not a chatbot bolted onto a dashboard, and it is not a forecasting black box. It is a frontier-LLM `tool-use loop` — Anthropic's **Claude Opus 4.7** and OpenAI's **GPT-5.5**, swappable per workload — that sits over the same governed panel everything else reads from. When you ask a question, the agent does not invent a number — it picks tools, runs SQL, calls a model endpoint, reads the result, and decides what to call next, until it has enough evidence to answer.

Architecture · one turn

READ-ONLY



THE CONTRACT

What the AI is allowed to do

- Read the panel through allow-listed tools.
- Compose tools — output of one is input to the next.
- Cite every number with the SQL or model endpoint that produced it.
- Refuse out-of-scope or out-of-evidence questions.

`It cannot mutate, cannot fabricate, cannot reach the open web.`

BRAIN

Claude · GPT

Claude Opus 4.7 & GPT-5.5 · tool-use APIs · streaming · iteration cap.

HANDS

18 tools

Datasets, models, spatial ops, scenarios — all declared with JSON Schema.

MEMORY

Panel

DuckDB-backed hex_panel + model outputs. The single source of truth.

Eighteen tools. Seven families. Every one declared, typed, and replayable.

The Canvas, the AI Analyst, and the Workbench all draw from the same registry. A tool is a contract: declared name, JSON-Schema input, typed output, supported scopes and modes, normalisation it accepts, and the data quality it reads. Add a tool to the registry and it appears, simultaneously, on the Canvas rail and in the AI Analyst's allow-list.

FAMILY	TOOLS	SCOPES
Datasets · Macro 5 tools	catalogue_browse · priority_surface · gdp_momentum · rank · compare	all
Labour & services 2 tools	service_pressure · equity_lens	muni · ward · hex
Property & land 2 tools	land_cover_change · transitions	muni · ward · hex
Spatial 2 tools	flow_explorer · morans_i	district · muni · ward
Trade & complexity 2 tools	analog_finder · opportunity_graph	muni · district · province
Scenarios 3 tools	trade_off_frontier · intervention_cockpit · annual_pack_studio	muni · district
Diagnostics 2 tools	data_quality_inspector · provenance_trace	all

A tool, declared

PRIORITY_SURFACE

```
// JSON-Schema, abbreviated
{
  "name": "priority_surface",
  "family": "datasets.macro",
  "scopes": ["national","province","district","muni","ward","hex"],
  "modes": ["twin","research"],
  "normalisation": ["local_diagnosis","national_comparable"],
  "reads": ["hex_panel","model_priority"],
  "input": {scope, year, weights?},
  "output": { surface, score, confidence, caveats }
}
```

WHY DECLARED

Surfaces refuse to render incoherent comparisons. A ward-only tool cannot be invoked at province scope.

WHY TYPED

The AI Analyst gets schema-validated outputs; broken contracts fail loudly at dev time, not in front of the MEC.

Every answer carries the receipts. Every query passes the gate.

An AI that can write SQL is an AI that can produce confident nonsense. We close that gap with four mechanisms — a SQL gate, a scope & normalisation contract, a caveats registry, and turn-by-turn replay. Together they turn "the AI said so" into "here is the SQL, here is the data, here are the caveats, here is the prior turn."

MECHANISM 01 · SQL GUARD

Allow-list before plan

```
sql_guard.check(query) →
  ✓ tables ∈ allow_list
  ✓ no DDL · no DML
  ✓ rows ≤ 100,000
  ✓ plan inspected · no Cartesian
```

MECHANISM 02 · SCOPE & NORMALISATION

Refuse incoherent comparisons

Every tool declares its scope. The model refuses to chart `native_raw` next to `national_comparable`. A ward tool cannot answer a province question.

MECHANISM 03 · CAVEATS REGISTRY

Every cell has a footnote

```
⚠ freshness    VIIRS ≥ 60d stale → flag
⚠ confidence   nowcast σ > threshold → flag
⚠ coverage     hex sample < 30 → flag
⚠ revision     StatsSA reweight pending → flag
```

MECHANISM 04 · REPLAY

Every turn is a fixture

Each turn captures the prompt, the tool calls, the SQL, the outputs, and the rendered answer. Re-run on a new model version and diff against the golden — regressions are caught before the room sees them.

Same panel. Same tools. Two modes, five workflow packs.

A municipal manager planning a refuse-collection rebid does not want the same first screen as a researcher exploring complexity scores. The system loads a **workflow pack** per role — same registry, same data, but a curated tool rail, a default Canvas layout, and a system prompt tuned to the user's question vocabulary.

Modes — the orientation

DEFAULT · TWIN

TWIN MODE

Decide

Curated rail. Defaults to muni or ward scope. Caveats inline. Annual-pack studio one click away.

priority_surface · gdp_momentum
service_pressure · intervention_cockpit
annual_pack_studio

RESEARCH MODE

Explore

Full registry. Hex scope unlocked. Diagnostics surface in every tile. Provenance always on.

all 18 tools · all 6 scopes
Moran's I · flow_explorer
provenance_trace · QA inspector

PACKS · 5

P1 · MUNICIPAL MANAGER

Twin · muni default · service-pressure dashboard · monthly review pack.

P2 · PROVINCIAL ECONOMIST

Twin · district default · GDP momentum · trade-off frontier.

P3 · INVESTOR / DFI

Research · analog finder · opportunity graph · resilience overlay.

P4 · RESEARCHER

Research · hex scope · Moran's I · custom SQL via guard.

P5 · EXECUTIVE BRIEFER

Twin · single-page board view · annual pack studio · auto-narrative with caveats.

A tile workbench. 18 tools. Two modes. Six scopes. One persistent layout.

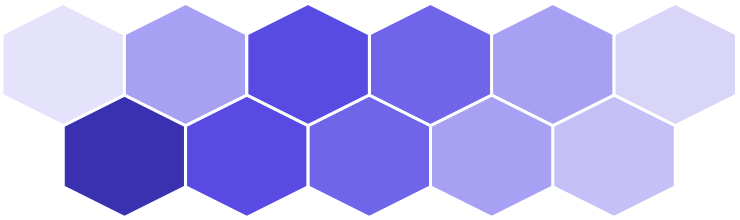
Where the dashboard answers a fixed set of questions, the **Canvas** is open — drop a tool, get a tile; rearrange tiles into the layout you need; pin, share, export. Each tile carries its own provenance and normalisation contract. The catalogue is a registry: every tool declares which modes, scopes, and normalisations it supports, so the surface refuses incoherent comparisons.

Canvas · 4-tile layout

MOCK

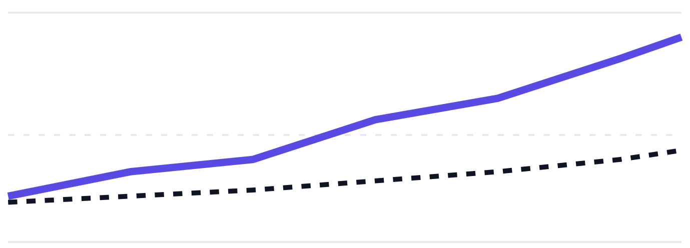
PRIORITY SURFACE · WARD

Investment-readiness



GDP MOMENTUM · DISTRICT

YoY trajectory



ANALOG FINDER · MUNI

Structurally similar

```
Tshwane      0.91
Ekurhuleni   0.87
Cape Town    0.79
Buffalo City 0.74
Mangaung     0.68
```

EVIDENCE INSPECTOR

Provenance · per tile

```
model_nowcasting · 2026-02
hex_panel · 57,330 rows
SEADsa GVA · 2013–2021
VIIRS nightlights · 2014–2024
conf 0.84 · freshness ✓
```

TOOL FAMILIES · 7 · 18 TOOLS

What's on the rail

- **Datasets · Macro** · catalogue browse, priority surface, GDP momentum, rank, compare
- **Property & land** · SANLC land-cover change, transitions
- **Trade · Scenarios** · analog finder, opportunity graph, trade-off frontier, intervention cockpit, annual pack studio
- **Labour & services** · service pressure, equity lens
- **Spatial** · flow explorer, Moran's I autocorrelation

MODES

2

Twin for municipal decisions; **Research** exposes the full grid.

SCOPES

6

National, province, district, muni, ward, hex.

NORMALISATION

3

`native_raw` · `local_diagnosis` · `national_comparable`.

A tile is a question. The Workbench is where you arrange them.

The Canvas is the surface; the **Workbench** is the verb. Drop a tool from the rail, configure once, and it becomes a tile — a self-contained, provenance-bearing visualisation at a fixed scope and normalisation. Tiles are first-class objects: pin them, duplicate them, link them so a click on one filters the others, share the layout as a URL, export the arrangement as a one-page brief.

Tile lifecycle

5 STATES

01 · DRAFT	02 · LIVE	03 · LINKED	04 · PINNED	05 · EXPORT
Pick & configure	**Tile renders**	**Cross-filter**	**Saved layout**	**One-page brief**
Choose a tool, set scope & normalisation. Preview before commit.	Tool runs against the panel. Provenance stamped on the tile footer.	Click a hex on tile A — tiles B, C, D scope to the same selection.	Layout becomes a named Canvas. URL-shareable. Belongs to a pack.	PDF · PPTX · annual pack. Caveats included. SQL hashes included.

WHAT LIVES ON A TILE

header	tool name · scope · year
body	map · chart · table · narrative
controls	scope-step · normalise · year-scrubber
evidence	SQL hash · sources · freshness
caveats	0..n flags from the registry
actions	pin · link · duplicate · export · ask AI

CROSS-FILTER

Brushing

Selecting a hex/ward on any tile broadcasts a scope-event; linked tiles re-render in place.

PERSISTENCE

URL state

Layout, scope, year, normalisation, filters — all in the URL. Send a link, get the same view.

ASK THE AI

Per-tile

"Explain this tile" hands the analyst exactly the SQL, the data, and the question.

EXPORT

Annual pack

A pinned Canvas becomes the seed for the next monthly or annual board pack.

WORKED CANVAS

"Where in Ekurhuleni should we site the next industrial incentive?"

Six tiles. One question. Cross-filtered. The user opens a fresh Canvas, drops priority_surface, and links four supporting tiles. The AI Analyst reads the same layout and writes the narrative. Total time, including export — under four minutes.

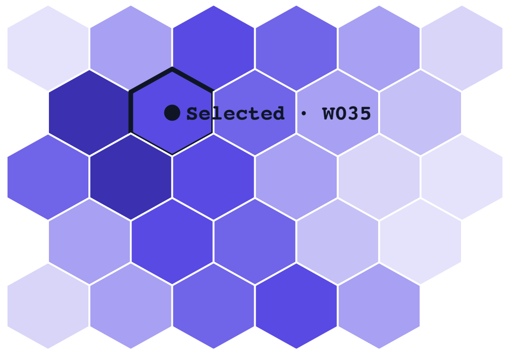
CANVAS · EKURHULENI · TWIN MODE

6 TILES · LINKED

Industrial incentive siting

A · PRIORITY_SURFACE · WARD

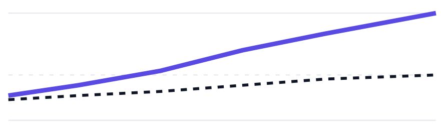
Investment-readiness · 2026



selection broadcasts → tiles B–F

B · GDP_MOMENTUM · W035

YoY · vs muni avg



+3.8 ppt YoY · conf 0.86

C · SERVICE_PRESSURE · W035

Composite index

0.74

⚠ above muni median (0.61)

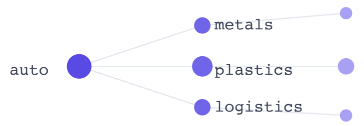
D · ANALOG_FINDER

Structural analogs

Pinetown 0.91
Mobeni 0.86
Hammarsdale 0.79

E · OPPORTUNITY_GRAPH

Adjacent sectors



F · AI ANALYST · NARRATIVE

"Why W035, in one paragraph"

W035 is in the top decile for investment-readiness, with momentum +3.8 ppt YoY and structural analogs in three operating industrial nodes (Pinetown, Mobeni). Service-pressure is elevated (0.74) — siting must couple with bulk-services upgrades. Adjacent opportunity space favours metals and plastics over light manufacturing.

tools: priority_surface, gdp_momentum, service_pressure, analog_finder, opportunity_graph · 5 calls · 1.8s

WHAT JUST HAPPENED

1. User dropped `priority_surface` at ward scope.
2. Selected hex W035 → broadcast to linked tiles.
3. Five tools auto-rendered at the selected scope.
4. "Explain this Canvas" → AI Analyst wrote tile F.
5. Pinned. URL shareable. Exported to PDF brief.

PROVENANCE · THIS CANVAS

hex_panel · 57,330 rows
model_priority · 2026-02 · conf 0.86
SEADsa GVA · 2013–2021
VIIRS nightlights · ≤ 7d
⚠ 1 caveat · service_pressure

TIME

3m 47s

From empty Canvas to exported brief.

REUSE

Saved as pack

Reopens next month with fresh data.

ECONOMETRIC LAB

Policy scenarios. Baseline vs adjusted. Uncertainty bands included.

§ 04 · SURFACE

A controlled environment for "what if" — increase EDIP rebates 20%, lose four weeks to load-shedding stage 6, expand the fault repair budget by R30m. The lab generates a baseline and adjusted trajectory with confidence intervals.

TARGETS
gva · employment

HORIZON
12 months · forward

ESTIMATOR
Panel · fixed effects · additive

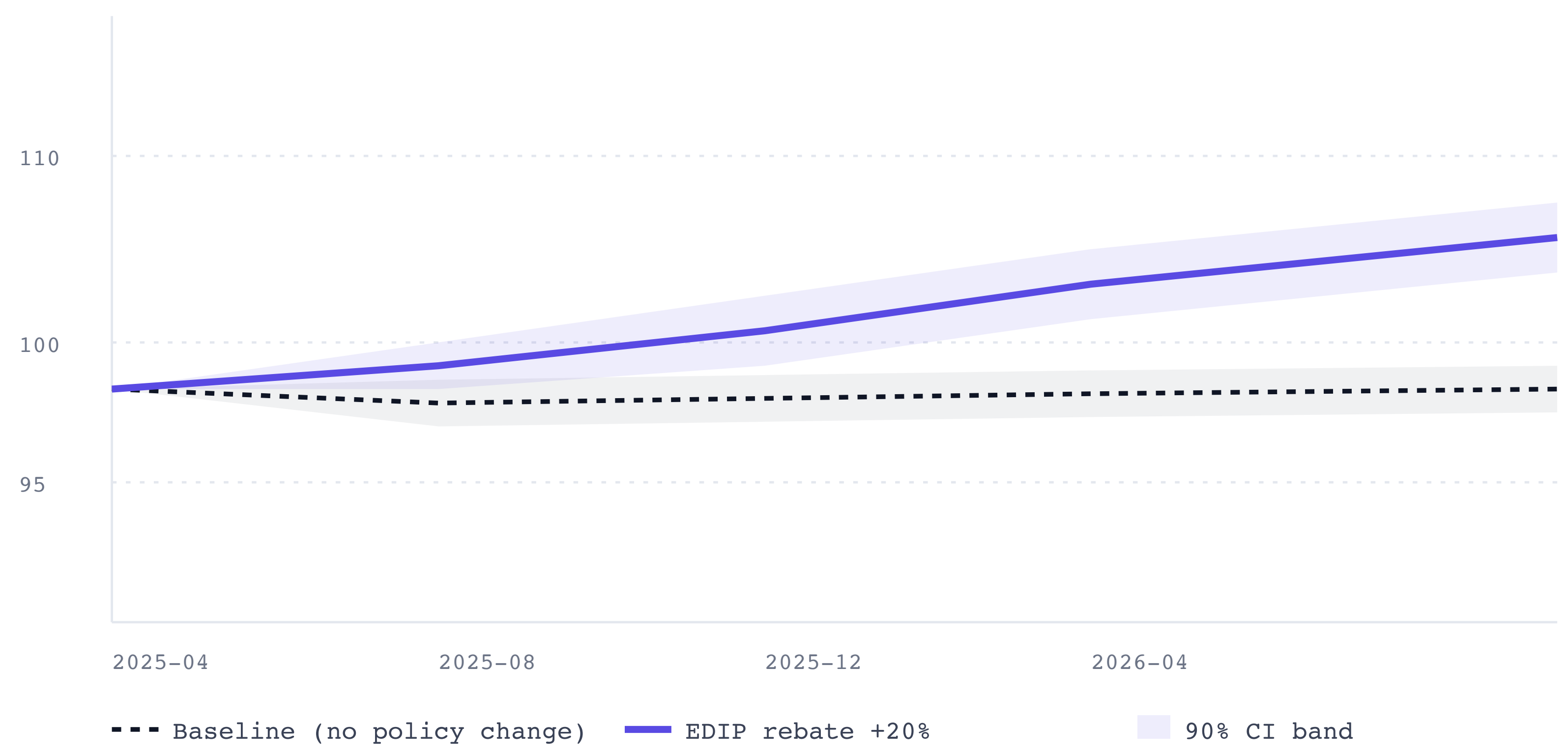
DIAGNOSTICS
R^2 · residual plots · placebo

API SURFACE

```
POST /api/v1/econometrics/run
GET  /api/v1/econometrics/coefficients
GET  /api/v1/econometrics/forecast
GET  /api/v1/econometrics/scenario-impact
GET  /api/v1/econometrics/diagnostics
```

Scenario · EDIP rebate +20%

12-MONTH FORECAST



WORKED EXAMPLE

From a question to a ranked list, in five steps.

QUESTION · IDP / SDF INPUTS · Q3 2025

"Which 25 wards should we prioritise for industrial CAPEX over the next planning cycle?"

STEP 01 · SCOPE

Translate to features

"Industrial CAPEX impact" → high `accessibility_index`, mid-to-high `eci`, elevated `fault_count`, and growing `growth_rate`.

STEP 02 · QUERY

Pull the panel

Single SQL on `hex_panel`, filter to last 24 months, project the four features per hex.

STEP 03 · SCORE

Composite ranking

Min-max each feature within the metro; weight per planning intent (default 0.35 / 0.30 / 0.20 / 0.15); aggregate to ward.

STEP 04 · VALIDATE

Cross-check

Top-25 ward list inspected against M5 resilience and M7 growth typology — flag any wards where the model would over-promise.

STEP 05 · DELIVER

Decision pack

Ranked CSV + interactive dashboard view + single-page PDF with assumptions, weights, and caveats.

Time elapsed, ingest to deliverable: **immediate** — a question of this shape runs against the current panel in real time, no rebuild required.

Reproducible. Tested. Deployable.

REPRODUCIBILITY

1 cmd

`make all-local` rebuilds the panel and all nineteen models without the external volume.

TEST SURFACE

1,059

Unit, integration, and validation tests. CI runs the full suite on every change.

DEPLOYMENTS

3

Local · Azure App Service · admin control plane (port 8100). Deployment runbook ships in `docs/deployment.md`.

SNAPSHOTTING

∞

`make snapshot-configs` writes timestamped config copies for rollback and review.

REST API · AUTH

X-API-Key on every route

Health is open. Everything else requires a key. Optional per-IP and per-key rate limits. Echoed X-Request-ID for tracing. Errors land in a single envelope shape.

PRIVACY

HMAC-SHA256 owner hashing

No MD5 fallback. Hashing key required at runtime. FTE values are masked under a per-hex small-cell threshold to prevent re-identification.

WHAT WE NEED FROM EKURHULENI

Three commitments to keep the platform live.

§ 06 · PROGRAMME

ASK 01

1

Nominated counterpart

A named, stable lead from your office — single point for Tier 4 data scoping, validation sign-off, and quarterly review.

TIME COMMITMENT · ~4 HRS / MONTH

ASK 02

4

Tier 4 onboarding workshop

One half-day with your data, finance, infrastructure, and economic-development leads to scope the Ekurhuleni operational layer (consumption, faults, valuation, incentives, firm registry).

OUTPUT · CO-DESIGNED TIER 4 SCHEMA

ASK 03

4

Quarterly review cadence

One 90-minute review per quarter. Walk through validation results, confirm the policy questions in flight, and re-prioritise the roadmap.

OUTPUT · SIGNED VALIDATION MEMO

In return: Tiers 1–3 of the platform refreshed against the Ekurhuleni backbone from day one; Tier 4 brought online together with full reproducibility and source code access — the same surface eThekwini runs today.

What we are watching, and what could break.

RISK	WHY IT MATTERS	MITIGATION	STATUS
FTE small-cell masking PRIVACY	Hexes with very few employees are masked to prevent re-identification. M1 calibration must work around the masked rows.	Calibration uses ward-level totals as anchor; downstream models tolerate masked inputs and surface them in the dashboard.	MITIGATED
External volume dependency /VOLUMES/SPATIALECON	GHSL rasters and SANLC live on the external volume. Without it, M7 and M8 cannot rebuild from raw.	Pre-built derived tables live in DuckDB. Pipeline runs `--skip-external` gracefully.	WATCHING
Anthropic API for chat / report export	The conversational extras and DOCX report export require credentials at request time. Non-AI routes are unaffected.	API startup no longer requires Anthropic credentials. AI routes degrade gracefully when the key is absent.	MITIGATED
Consumption feed lag	Utility consumption data arrives with a 2–6 week lag. Recent-month nowcasts widen confidence bands.	Dashboard surfaces freshness per variable; recent-month estimates explicitly tagged as provisional.	WATCHING
Calibration window drift	Long-running structural shifts (load-shedding regime, hybrid work) can detune M1's proxy regression.	Retraining cadence is quarterly; validation harness fails the build if M1's r^2 drops below threshold.	MONITORED
Methodology novelty M2 AT SUB-CITY SCALE	Hidalgo-Hausmann at intra-city resolution is novel. Peer-review and validation surfaces are still maturing.	Face-validity checks (V2) plus a published methodology paper from the research loop.	RESEARCH

ROADMAP

Two horizons. 90 days. 12 months.

HORIZON 1 · 90 DAYS

Pilot package — sign-off & embed

- Confirm validated nowcast for 2024-Q4 against ward GDP
- Land 2 pilot policy questions through the Econometric Lab end-to-end
- Stand up read-only dashboard access for IDP / SDF planning leads
- Sign quarterly review memo template and confirm cadence
- Lock the consumption-feed contract and SFTP path

OWNER · SPATIAL ECONOMICS + NOMINATED COUNTERPART

HORIZON 2 · 12 MONTHS

Commercial package — operate & extend

- Production deployment on Azure App Service with admin control plane
- Quarterly model re-training; published validation memos
- M9 — service-delivery urgency index (in development)
- National lattice: bind Ekurhuleni outputs into the existing eThekwini pilot + Western Cape urban-growth-monitor cohort
- Public-facing leave-behind report (annual)

TRIGGER · PILOT SIGN-OFF + SIGNED COMMERCIAL-PACKAGE SOW

SPATIAL ECON | CLOSING

NEXT WORKING SESSION

Sub-city. Monthly. Validated.

eThekwini already runs the live, validated slice of the platform — built once, tested 1,059 ways, queried hundreds of times. Ekurhuleni can have the same surface from the national backbone today; the next working session co-designs the Tier 4 data feed, names the counterpart, and locks the review cadence so the platform runs as a service, not a project.

ACTION 01	Confirm nominated counterpart by 2026-05-09
ACTION 02	SFTP path + monthly cadence locked by 2026-05-16
ACTION 03	First quarterly review · 2026-Q3

CONTACT & HANDOVER

Let's build.

All material in this deck reflects the v1.0 pilot package. Code, validation, and the full data dictionary are available on request.

EMAIL	insights@spatialecon.co.za
CODE	EThekwini-Spatial-Economic-Model
VERSION	v1.0 · 2026-04-27